

SIMATS ENGINEERING
ASSESSMENT TOOL 2
Case study

COURSE NAME: Computer Networks

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

Code of Conduct:

I G. Santosh Kumar (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

G. Santosh Kumar

35
40

Total Marks:40

QUESTIONS:4

1.A multinational IT company uses a cloud-based video conferencing platform for daily client meetings. During peak business hours, employees experience frequent voice distortion, frozen video frames, and delayed communication. Network monitoring reports indicate an **8% packet loss, 180 ms jitter**, and fluctuating bandwidth utilization. The IT administrator suspects that the issue is related to the transport protocol or application-level buffering strategy.

Questions

1. Analyze whether the problem is primarily caused by the **transport protocol (TCP/UDP)** or the **application layer**.
2. Justify your answer based on the communication requirements of real-time multimedia applications.
3. Recommend suitable protocol-level or application-level improvements to enhance conferencing quality.

2.A multinational software company hosts its web services across multiple continents. Employees frequently report long delays before the application homepage loads. Network analysis shows that the **average DNS lookup time is approximately 800 ms**, significantly affecting user experience. The organization plans to redesign its DNS infrastructure while ensuring continuous service availability during server failures.

aQuestions

1. Analyze the reasons behind the high DNS resolution time.
2. Propose a suitable DNS architecture using techniques such as **Anycast DNS, DNS pre-fetching**, and **TTL optimization** to reduce the lookup time below **50 ms**.
3. Evaluate the advantages and limitations of your proposed architecture in terms of performance, scalability, and availability.

3.A cloud service provider delivers applications to millions of users worldwide. During promotional events, users experience intermittent delays in accessing cloud-hosted services. Investigation reveals that DNS queries are taking nearly **800 ms** because all requests are directed to a single primary DNS server. The company wants a highly available DNS solution capable of handling global traffic efficiently.

Questions

4. Examine the impact of centralized DNS architecture on application performance.
5. Design a distributed DNS solution that minimizes lookup latency while maintaining high availability.
6. Analyze how **TTL values**, **Anycast routing**, and **DNS caching** influence system performance and fault tolerance.

4.A financial institution operates an electronic stock trading platform where thousands of buy and sell orders are submitted every second. During market opening hours, the **average order acknowledgment latency increases from 1 ms to 40 ms**, resulting in delayed trade confirmations and customer dissatisfaction. Network engineers observe increased TCP retransmissions, longer connection establishment times, and excessive buffering.

Questions

1. Analyze three possible TCP-related factors responsible for the latency increase, considering **flow control**, **Nagle's algorithm**, and **SYN queue limitations**.
2. Explain how each factor affects transaction processing in high-frequency systems
3. Recommend appropriate TCP configuration changes to reduce latency and improve transaction reliability.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions	Logical solutions with adequate	Basic solutions with weak	Incorrect or no solution provided

Video Conferencing - TCP (UDP) & Application layer.

Analyze whether the problem is primarily caused by the transport protocol or application layer.

The problem is primarily related to the transport protocol & network conditions, especially the high 8% packet loss & 180 ms jitter. Real-time video conferencing normally uses UDP because it avoids retransmission delays.

However, application-level buffering can also affect the quality when packets arrive late or out of order.

2. Justification

Real-time multimedia applications require low latency, low jitter, & continuous data delivery. TCP provides reliable delivery through acknowledgment & retransmissions, but retransmitting lost packets can introduce additional delay. Therefore, UDP is more suitable for voice & video. However, high packet loss & jitter can still cause voice distortion, frozen frames, & delayed communication.

3. Improvements

The organization should use UDP with RTP (RTCP, etc.). QoS should be configured to prioritize audio & video traffic. Increasing bandwidth & reducing congestion & packet loss will further improve conferencing quality.

2. DNS optimization

1. Reasons for high DNS resolution time

The 800 ms DNS lookup time may be caused by geographically distant DNS servers, network congestion, overloaded servers, poor DNS caching & multiple DNS resolution steps. A centralized DNS infrastructure can also increase delay for users located far from the server.

2. proposed DNS architecture

The company should deploy anycast DNS servers. In multiple geographical regions. Anycast directs users to a nearby DNS server, reducing network latency. DNS caching & pre-fetching should be used for frequently accessed domains. Suitable TTL values should keep records cached for an appropriate period & reduce repeated lookups.

Advantages & Limitations

This architecture provides lower latency, scalability, reliability, & fault tolerance. If one DNS server fails, requests can be redirected to another server. However, anycast requires additional infrastructure & monitoring. High TTL values may also result in stale DNS records.

Impact of DNS

Impact of centralized DNS

A centralized DNS architecture sends requests to a single primary server. During promotional events, the large number of requests can overload the server & increase DNS lookup time to around 800ms. This delays application access & creates a single point of failure.

5. Distributed DNS solution

The company should deploy multiple DNS servers across different geographical regions using AnyCast routing. Users are directed to a nearby DNS server. DNS Caching, load balancing, & redundant servers should also be used. This distributes the workload & reduces lookup latency.

6. Effect of TTL, AnyCast & caching

TTL determines how long DNS records remain cached. A suitable TTL reduces repeated DNS queries & server load. AnyCast routing directs users to the nearest available DNS server. DNS caching allows frequently requested records to be answered quickly without contacting the authoritative server. These techniques improve performance, scalability & fault tolerance.

4. stock trading platform - TCP latency

They are three types

1. flow control:- TCP uses the receiver's window to control data transmission. If the receiver cannot process data quickly, the sender may have to wait.

2. Nagle's algorithm:- It combines small TCP packets to reduce network overhead. However, this can introduce delays for small, time sensitive trading messages.

3. SYN queue limitation:- During market opening, a large number of connection requests can fill the server's SYN queue, causing delays in establishing new connections.

2. Effect of transaction processing

flow-control delays can slow the transmission of trading messages. Nagle's algorithm can delay small buy & sell orders. SYN queue limitations. Increase connection establishment time. together these factors can increase order acknowledgement latency from milli-seconds to much bigger values.

Commanded Improvements

uses TCP-NODELAY for latency-sensitive connections where appropriate. Tune TCP send & receive buffers to avoid unnecessary flow-control delays. Increase the SYN backlog to handle large numbers of connections used to avoid repeated handshakes. These changes can reduce latency while maintaining TCP's reliable delivery.

